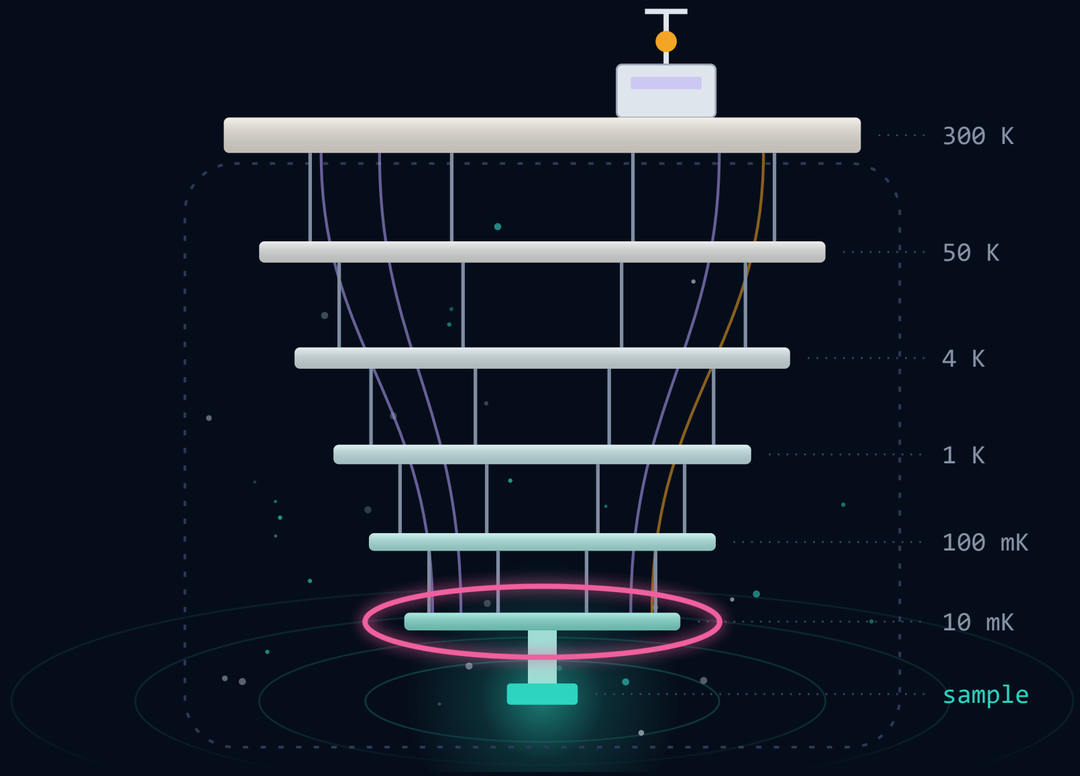
VIBECHECK · THE USABLE FORMAL METHODS HACKATHON · SAN FRANCISCO · NOV 2026

CryoLean

Find the failure before the cooldown, not at 10 mK.

heat budgets | cooldown schedules | Lean 4 proofs

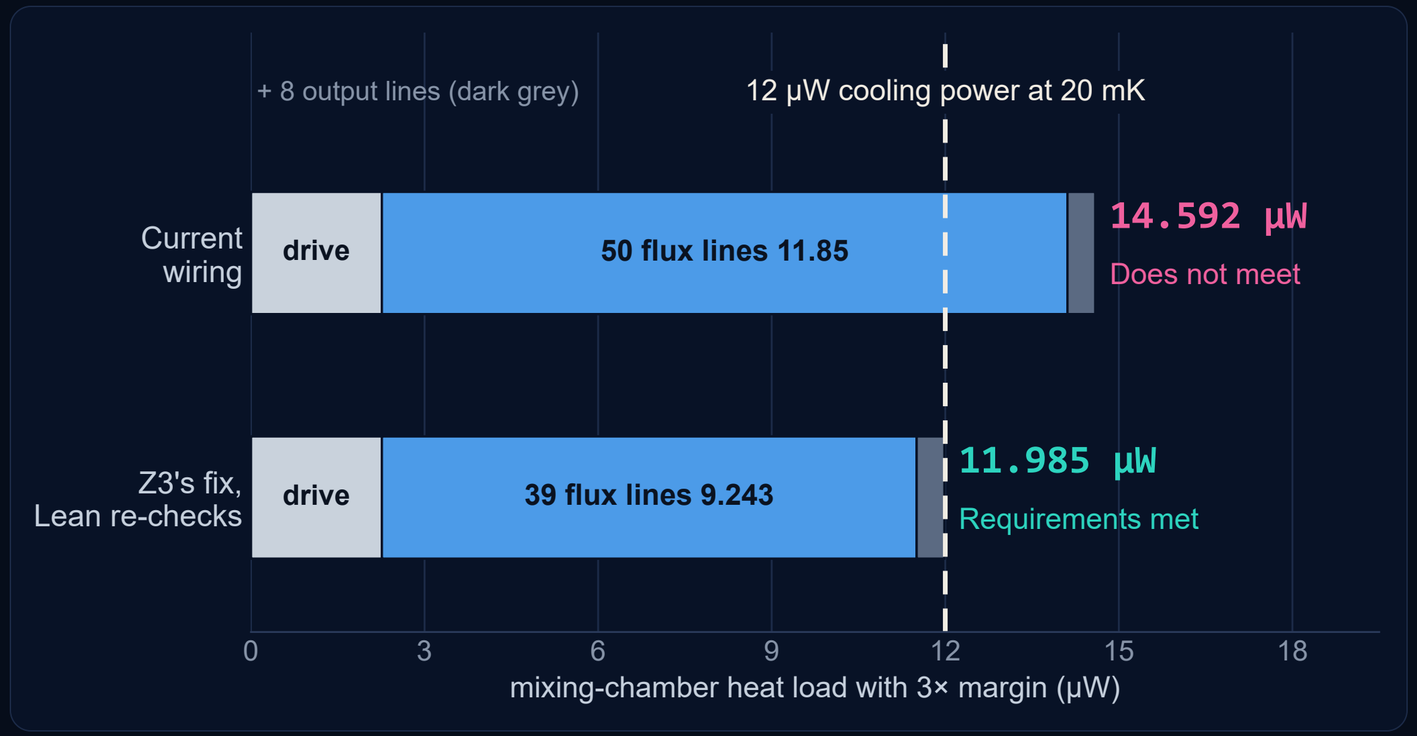


Hanah Rahman · Nicholas Brandenburg

hrahman12.github.io/CryoLean · github.com/hrahman12/CryoLean · generic cryostat drawn in code, no real model

/ COUPLED PHYSICS · 50-QUBIT WIRING

Each check passes. Together they fail.



10 ≤ 20 mK ✓
base temperature

168 ≥ 124 ✓
wiring lines

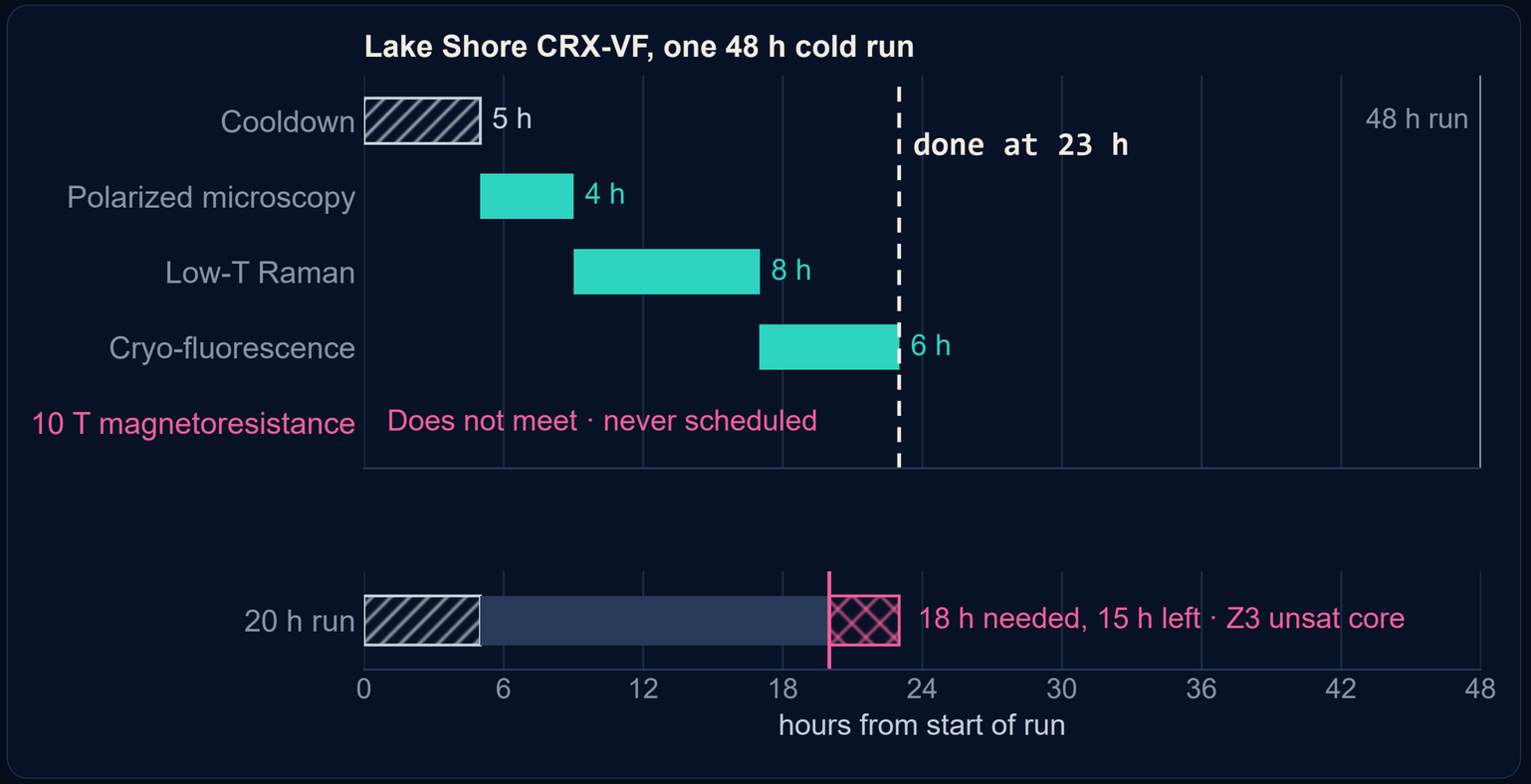
14.592 > 12 µW ✗
heat from all lines together

Bluefors LD400 mixing chamber. Z3's fix leaves **15 nW** of slack.

docs/examples/demo_b_coupled.json · paper/data/e5_heat.json, fig_heat_budget.json · line loads from Krinner et al., EPJ Quantum Technology 2019. Honest limit: with an assumed ±20 % per-line uncertainty the 39-line design passes about 51 % of the time; 95 % needs at most 33 lines (paper/data/fig_uncertainty.json).

/ SCHEDULING · ONE COLD RUN

One cooldown, three experiments, a proved schedule.



23 h

to finish all three accepted experiments in one cold run

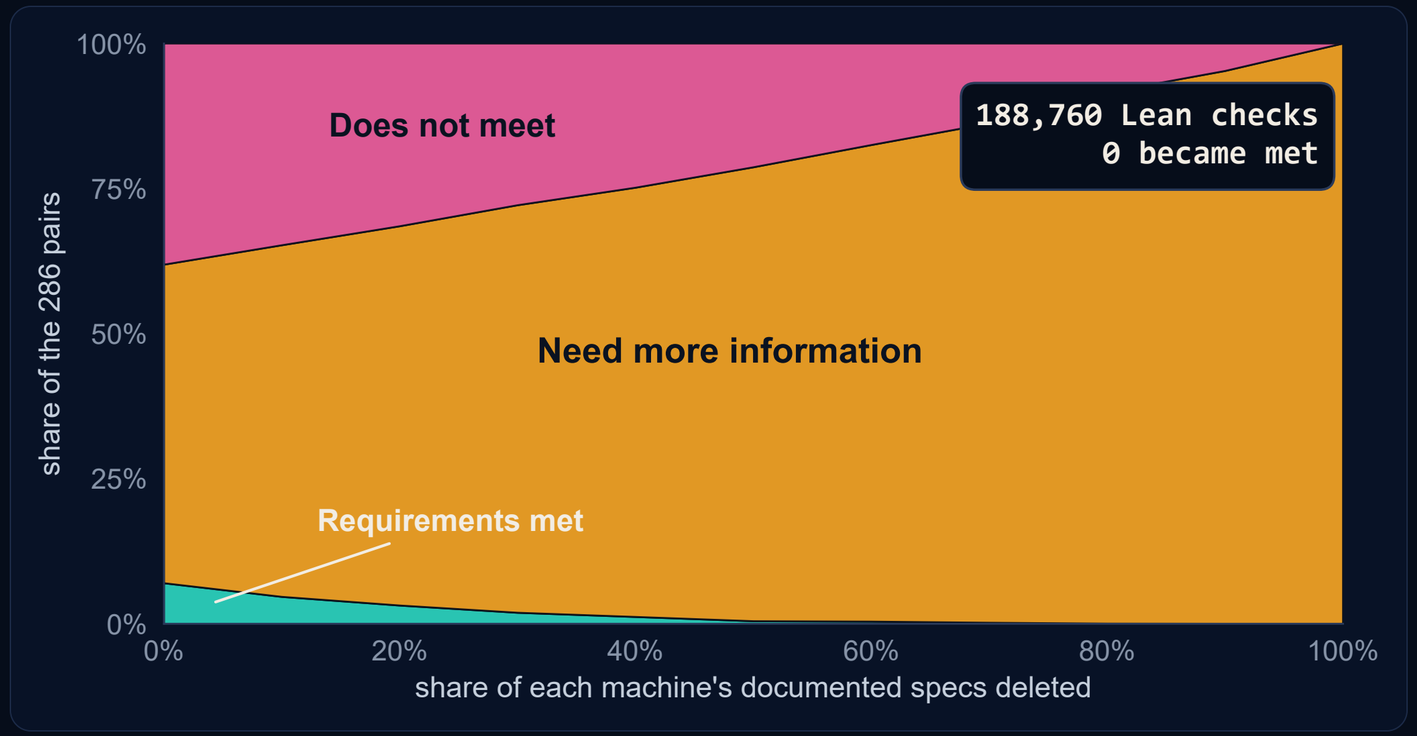
400 / 400

tampered schedules where Lean agrees with an independent checker

docs/examples/demo_d_campaign_schedule.json · paper/data/e6_schedule.json (139 valid, 261 rejected) · Lean proves the schedule valid, not minimal. The 20 h case is Z3's unsat core, not a Lean proof.

/ WHAT WE DON'T KNOW

Missing data never becomes a pass.



665

of 2,832 capability entries stored as UNKNOWN, not guessed

verdict_refines

Lean theorem: new data can only resolve Need more information

Delete specs, and verdicts only drift toward **Need more information**.

paper/data/fig_incomplete_info.json (paper evaluation: 286 pairs, 30 random deletions per level) · data/cryostats/*.json · lean/CryoLean/Monotone.lean. Proofs cover our model and the sourced data, not the machine.